

Ex: $L = \{a^n b^n c^n \mid n \geq 0\}$ is not a CFL.

Assume L is a CFL.

Then L must follow pumping lemma.
Suppose pumping length is p .

$$S = a^p b^p c^p, |S| = 3p \geq p$$

Then by pumping lemma, S must be able to be decomposed into 5 components where $S = uvxyz$, s.t

1. $uv^i xy^i z \in L, \forall i \geq 0$

2. $|v| > 0$

3. $|vxy| \leq p$

$$S = \underbrace{a a a \dots a}_{p \# \text{ of } a\text{'s}} \underbrace{b b b \dots b}_{p \# \text{ of } b\text{'s}} \underbrace{c c c \dots c}_{p \# \text{ of } c\text{'s}}$$

Case 1: vxy contains the same set of characters. Then v and y can only be allocated to the same type of character. Therefore, when we pump uv^ixy^iz , the # of one type of characters increases compared to other two types of characters.

$$\therefore uv^ixy^iz \notin L$$

Case 2: vxy contains mix of characters.

Subcase 2.1, v and y contain the same character. We handled this in case 1.

$$\text{ex: } vxy = \text{aaa} \dots \text{a bbb} \dots \text{b}$$

$$v = \text{aaa} \dots \text{a} \quad y = \text{aaa} \dots \text{a}$$

Subcase 2.2 v and y contain two different character types.

$$\text{ex: } vxy = \text{bbb} \dots \text{b ccc} \dots \text{c}$$

$$v = \text{bbb} \dots \text{b} \quad y = \text{ccc} \dots \text{c}$$

Now, when we pump up, uv^ixy^iz contains different number of type 3 character.

$$\therefore uv^ixy^iz \notin L$$

Subcase 2.3 either v or y contains a mix of characters.

$$vxy = aaa \dots a bbb \dots b$$

$$v = aa \dots ab \dots b \quad y = bb \dots b$$

Now, when we pump up, $uv^i xy^i z$ contains different order of a, b , and c 's.

$$\therefore uv^i xy^i z \notin L$$

Now, we considered all the possible decompositions for s , and none of them are pumpable.

$\therefore L$ is not a CFL.

$C = \{ ww \mid w \in \{0,1\}^* \}$ is not a CFL.

Assume C is a CFL.

Then C must follow pumping Lemma.

Suppose pumping length is p .

Since $|S| \geq p$, s must be able to be decomposed into $s = uvxyz$ s.t

1. $uv^i xy^i z \in C, \forall i \geq 0$

2. $|v| > 0$

3. $|vxy| \leq p$

case 1: vxy does not straddle the mid point of s .

In this case $uv^2 xy^2 z \notin C$.

The reason is that, in $uv^2 xy^2 z$, the first character of string 1 and first character of string 2 does not match.

case 2: If vxy crosses mid point, consider $uv^0 xy^0 z = uxz = o^p i^j o^j i^p$, where $i, j \neq p$.

$\therefore uv^0 xy^0 z \notin C$.

$\therefore C$ is not a CFL.



Prove $L = \{a^n b^j \mid n = j^2\}$ is not a CFL.

Assume L is a CFL.

Then, let p be the pumping length.

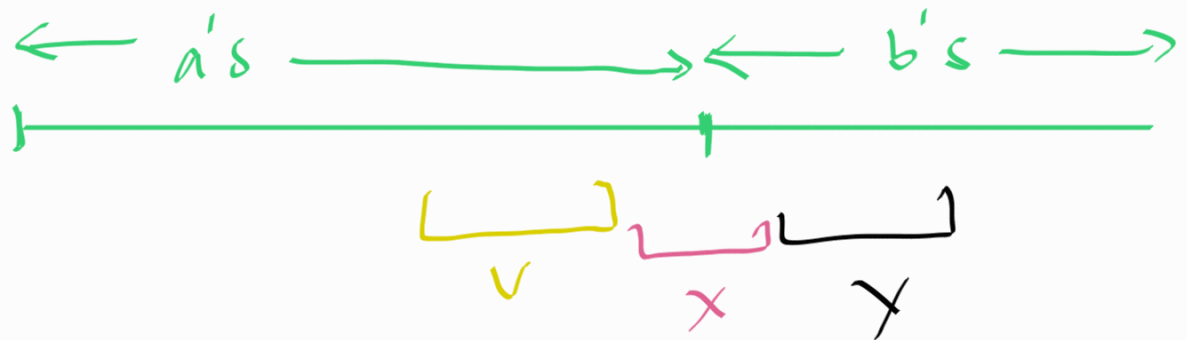
Let $s = a^{p^2} b^p$

$S = \underbrace{a a a a \dots a}_{p^2 \# a's} \underbrace{b b b b \dots b}_{p \# b's}$

Case 1: Either v or y crosses the breakpoint. Then, when pumping up, we will have out of order a 's and b 's.

$\therefore uv^i xy^i z \notin L$

case 2': Both v and y does not cross the break point.



k_1 # of a 's to v

k_2 # of b 's to y

consider uv^ixy^iz .

$$|uv^ixy^iz| = p^2 + p + (i-1)k_1 + (i-1)k_2$$

subcase 2.1 $(k_1 = 0) \wedge (k_2 = 0)$

This violates condition 2.

subcase 2.2 $(k_1 = 0) \vee (k_2 = 0)$

then $uv^0xy^0z \notin L$ since this would imbalance the # of a 's and # b 's.

$\therefore uv^0xy^0z \notin L$.

subcase 2.3 $(k_1 \neq 0) \wedge (k_2 \neq 0)$

consider $uv^i x y^i z$.

Now pumpdown, $i=0$

$$|uv^0 x y^0 z| = p^2 + p - k_1 - k_2$$

Suppose $uxz \in L$

$$p^2 - k_1 = (p - k_2)^2$$

$$p^2 - k_1 = p^2 - 2pk_2 + k_2^2$$

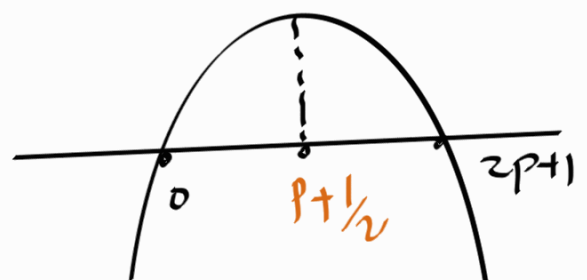
$$k_1 = -k_2^2 + 2pk_2$$

I am going to consider $|vxy|$

$$|vxy| = k_1 + k_2 + |x|$$

$$= \underbrace{-k_2^2 + 2pk_2 + k_2}_{f(k_2)} + |x|$$

$$f(k_2) = -k_2^2 + k_2(2p+1) = -k_2(k_2 - (2p+1))$$



$$f'(k_2) = -2k_2 + (2p+1)$$

$$f'(k_2) = 0 = -2k_2 + 2p+1, \quad k_2 = \frac{2p+1}{2} \\ = p + \frac{1}{2}$$

This means from $k_2=1$ to p $f(k_2)$ increases.

$$f(k_2) = -k_2^2 + k_2(2p+1)$$

$$f(1) = -1 + 2p+1 = 2p$$

$$f(p) = -p^2 + p(2p+1) = -p^2 + 2p^2 + p \\ = p^2 + p$$

Here, we show $f(k_2)$ which is the length of $|vxy|$

$$\therefore \forall k_2 \in [1 \dots p] : f(k_2) > p$$

$\therefore$ There is no decomposition of $uvxyz$ which is pumpable.

$\therefore L$ is not a CFL.

